PROBLEM STATEMENT :-

8-BIT INCREMENTER

INTRODUCTION

- An 8-bit incrementer is a digital circuit that adds 1 to an 8-bit binary number. It's commonly used in digital counters, processors, and address generation in computers.

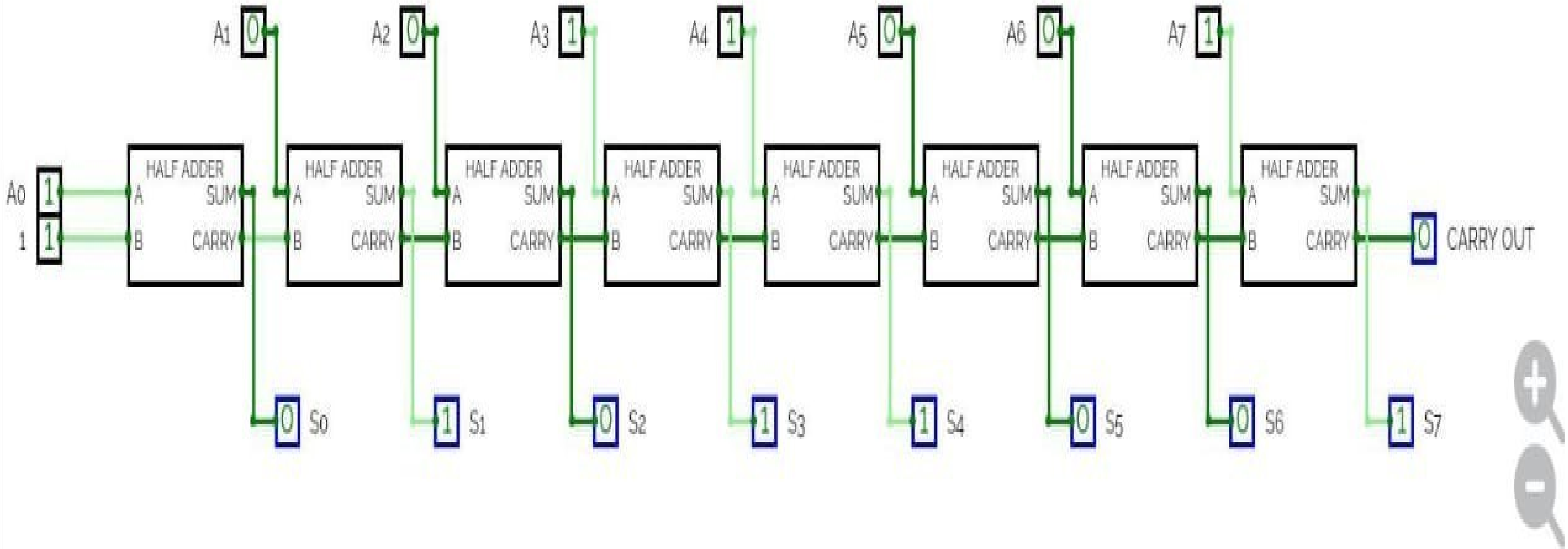
INCREMENTER CONCEPT

- An incrementer is a digital circuit that increases a binary number by one. In an 8-bit incrementer, each increment causes a binary number to increase from 00000000 to 00000001, and so on.

LOGIC DESIGN OF AN 8-BIT INCREMENTER

- An 8-bit incrementer can be built using a series of 8 half adders, each adding a carry-in from the previous bit. Only the first bit's carry-in is set to 1.

LOGIC DIAGRAM OF 8 BIT INCREMENTER



THEORY

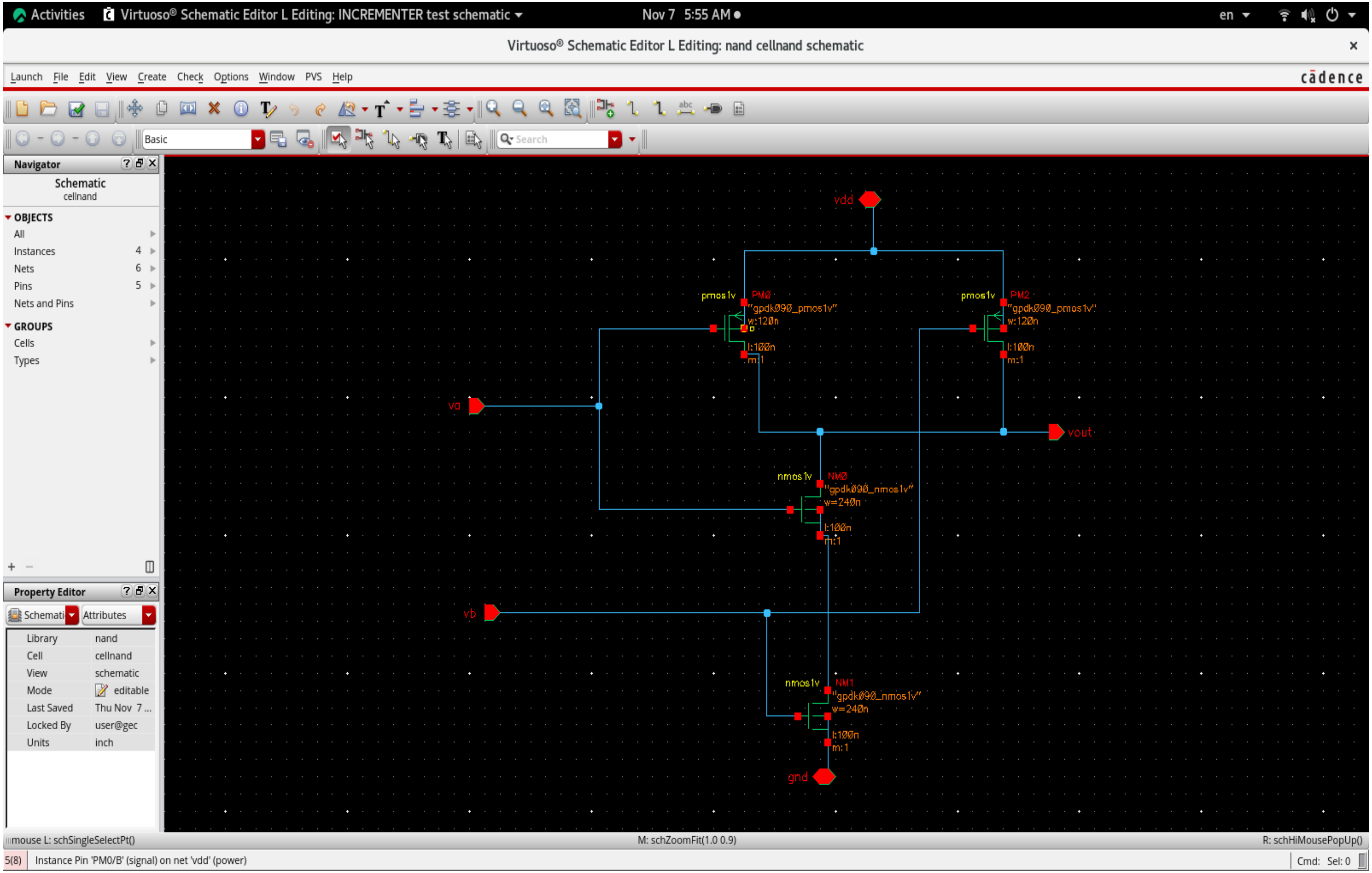
One of the inputs to the least significant half-adder is connected to logic-1 and other input is connected to the least significant bit of the number to the incremented.The output carry from one half-adder is connected to one of the inputs of the next-higher order half adder,The output carry from one half-adder is connected to one of the inputs of the next-higher order half-adder.The ciruit receives the 8 bits from A0 TO A7, adds one to it, and generated the incremented output in S0 through S7. The output carry C7 will be 1 only after incremening binary 11111111. This also causes outputs S0 through S7 to go to 0.

TRUTH TABLE

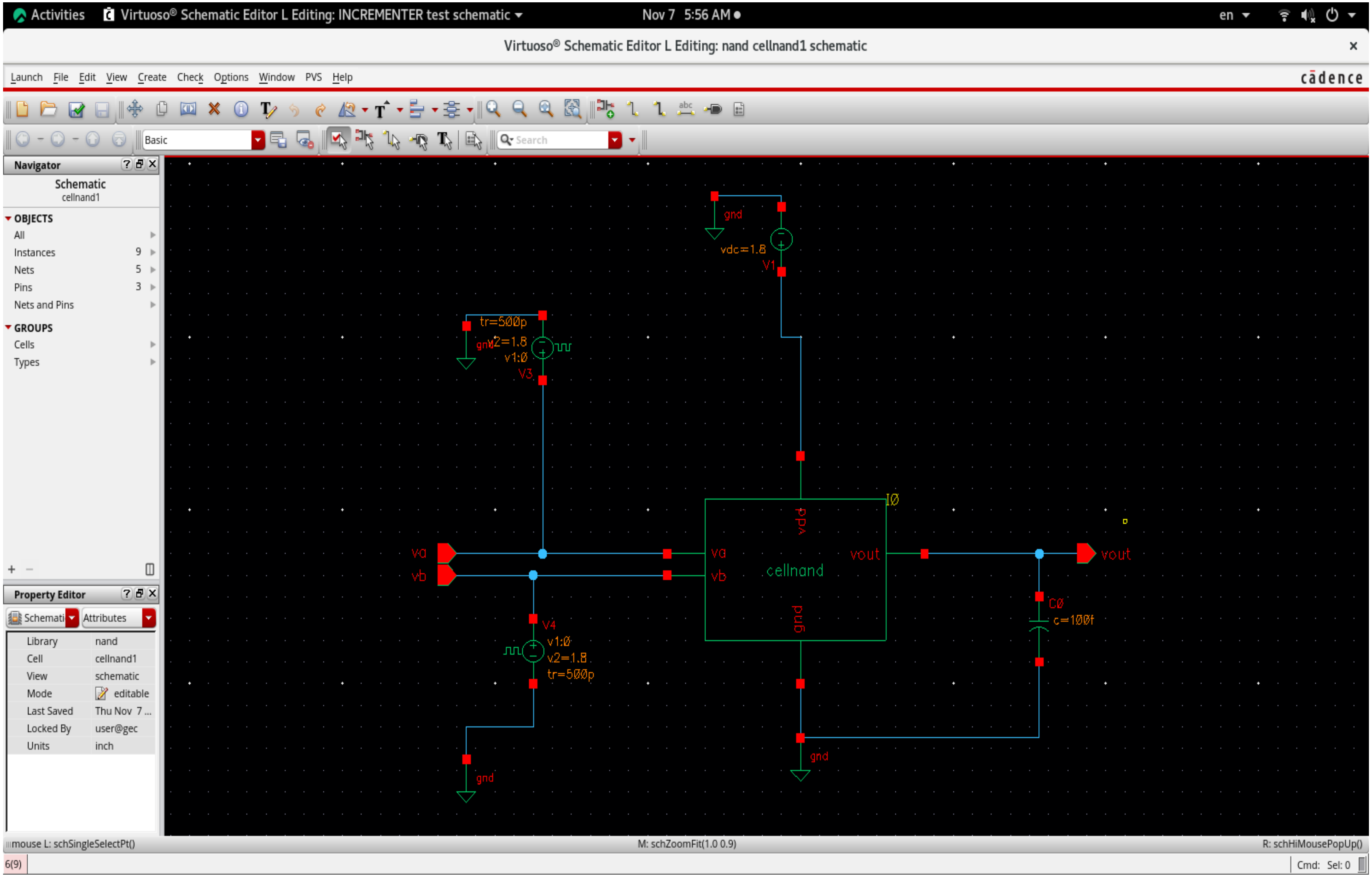
- The truth table for an 8-bit incrementer shows each binary input and the resulting incremented output.
- Example:
- Input: 00000001 -> Output: 00000010
- Input: 11111111 -> Output: 00000000

(carry is generated which results in overflow)

STEP 1 :- MAKE NAND GATE IN CADENCE USING PMOS AND NMOS.

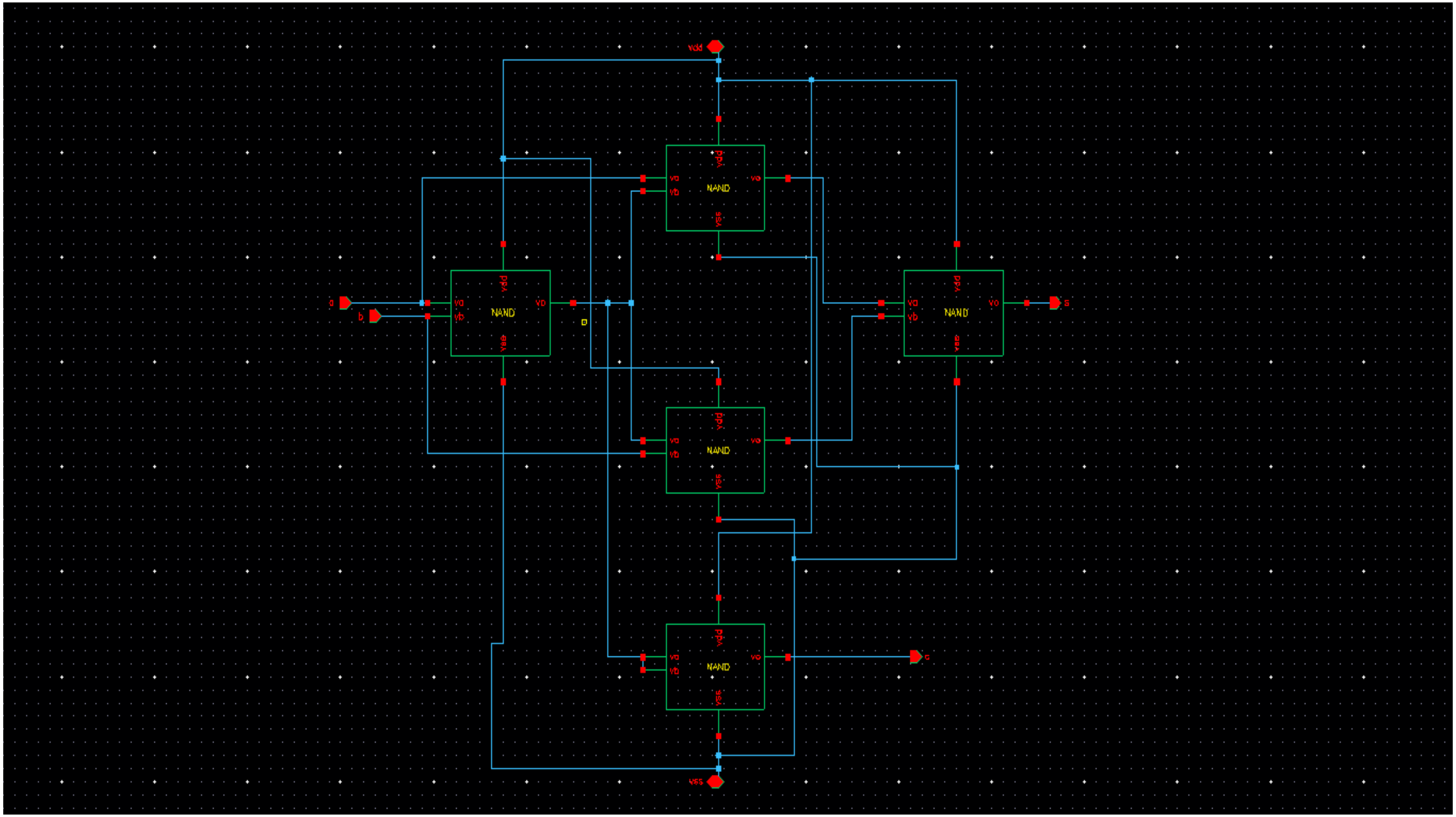


NAND GATE USING SYMBOL.

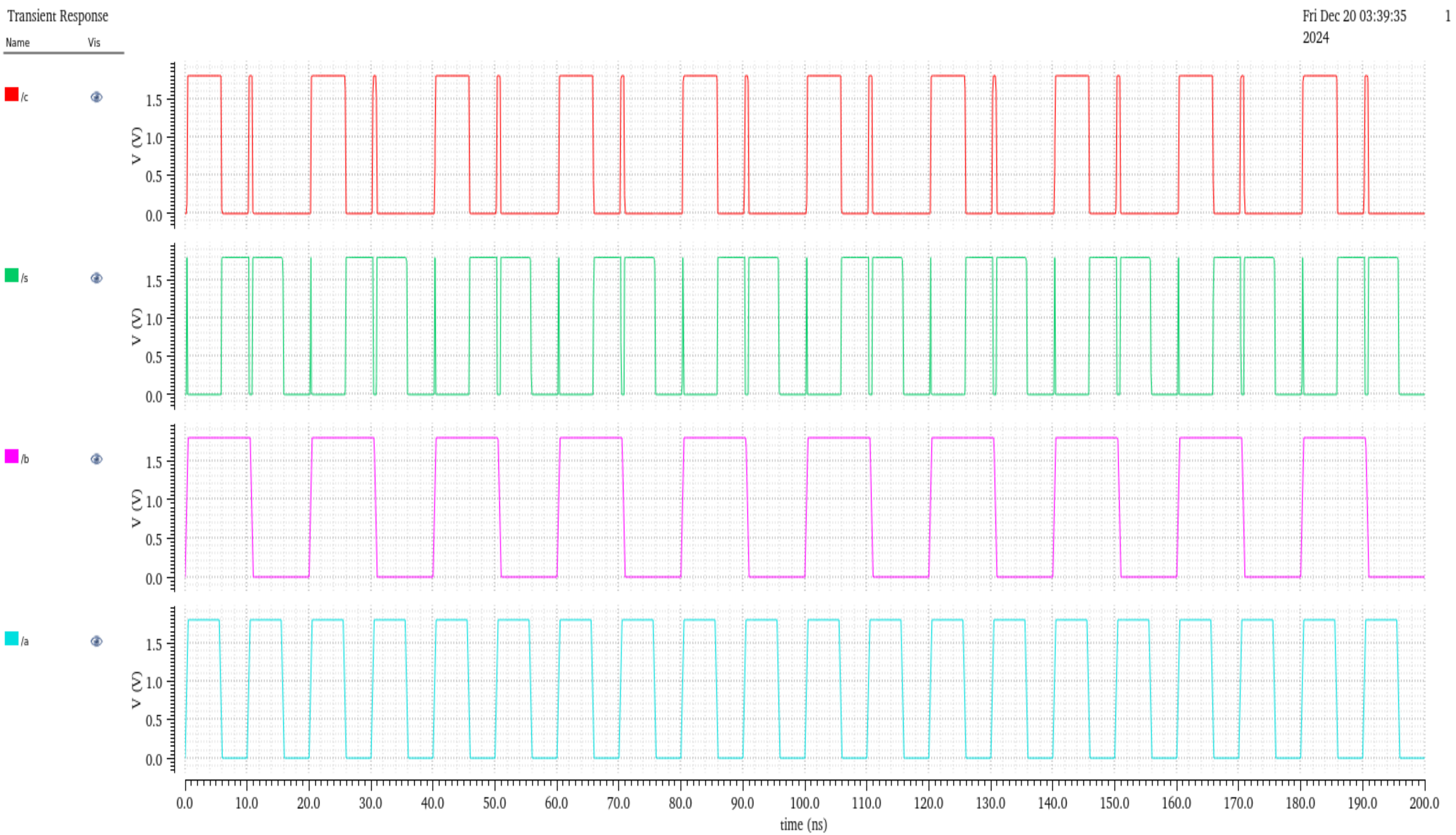


OUTPUT WE GOT FOR NAND GATE.

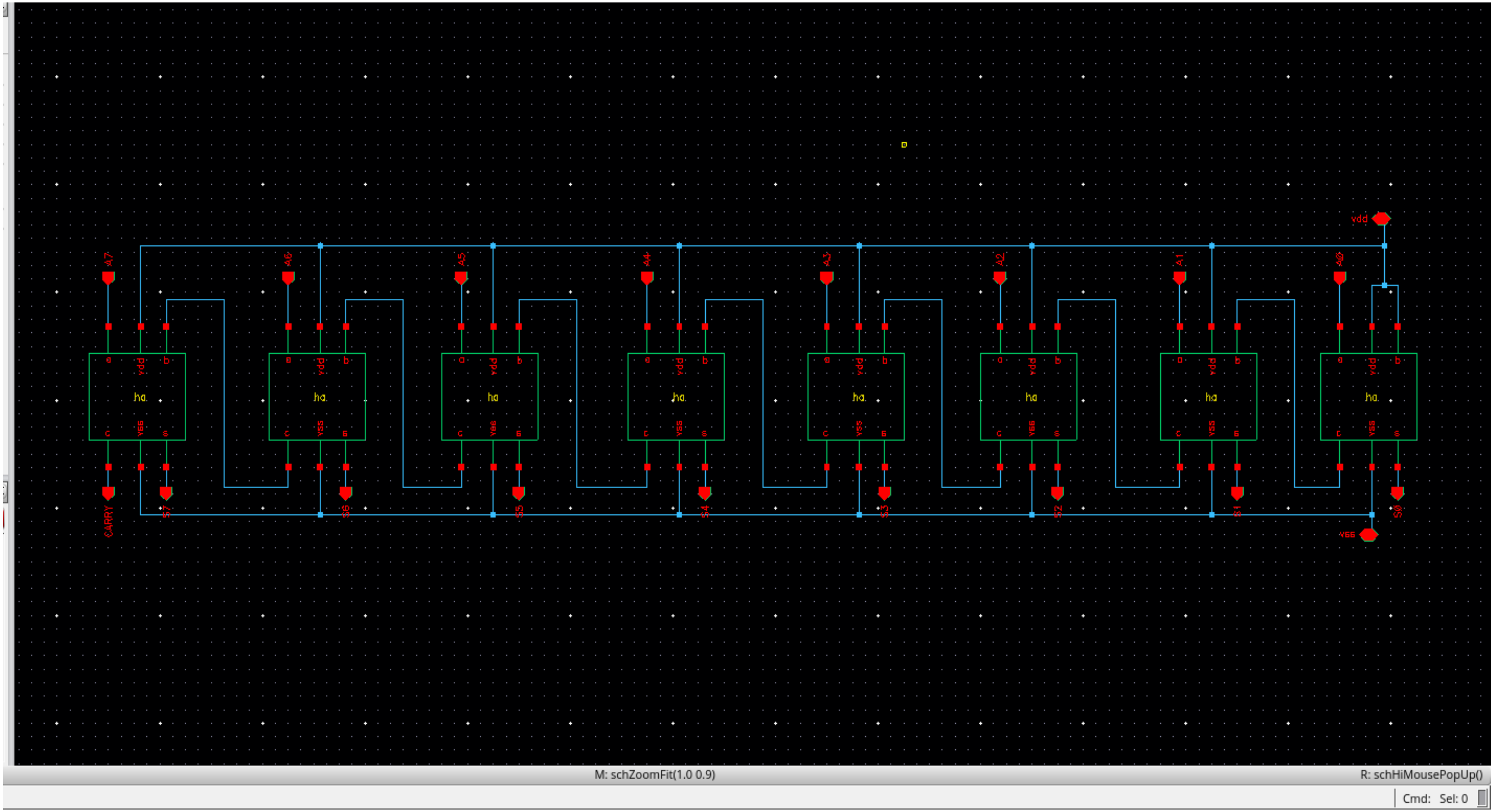
STEP 2:- MAKE HALF ADDER USING NAND GATES.



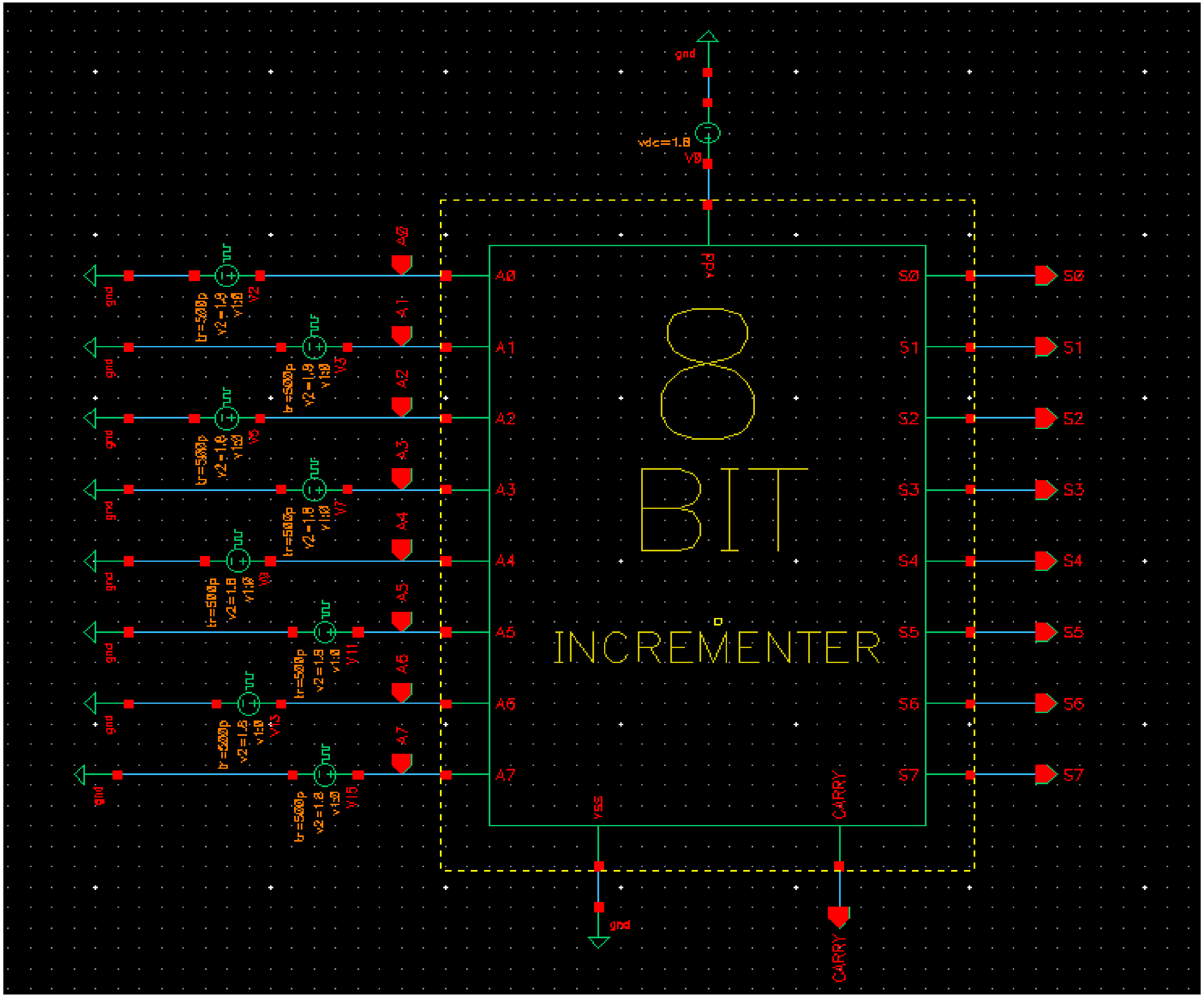
OUTPUT WE GOT FOR HALF ADDER.



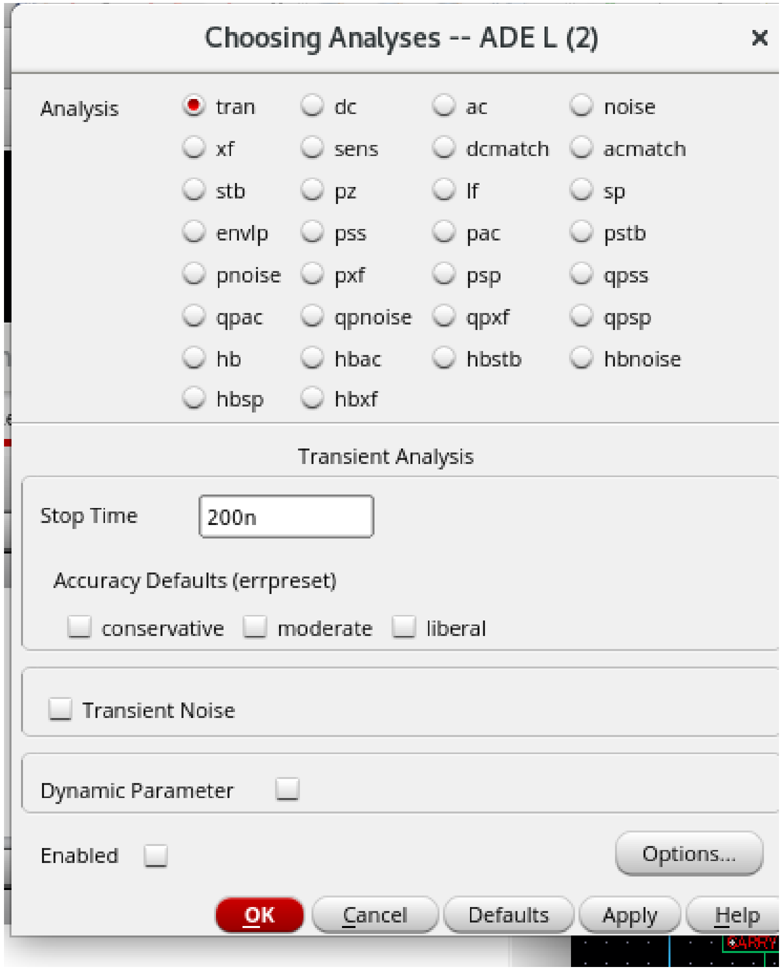
STEP 3 :- MAKE 8-BIT INCREMENTER USING 8 HALF ADDERS.



CREATE A SYMBOL FOR 8 BIT INCREMENTER.



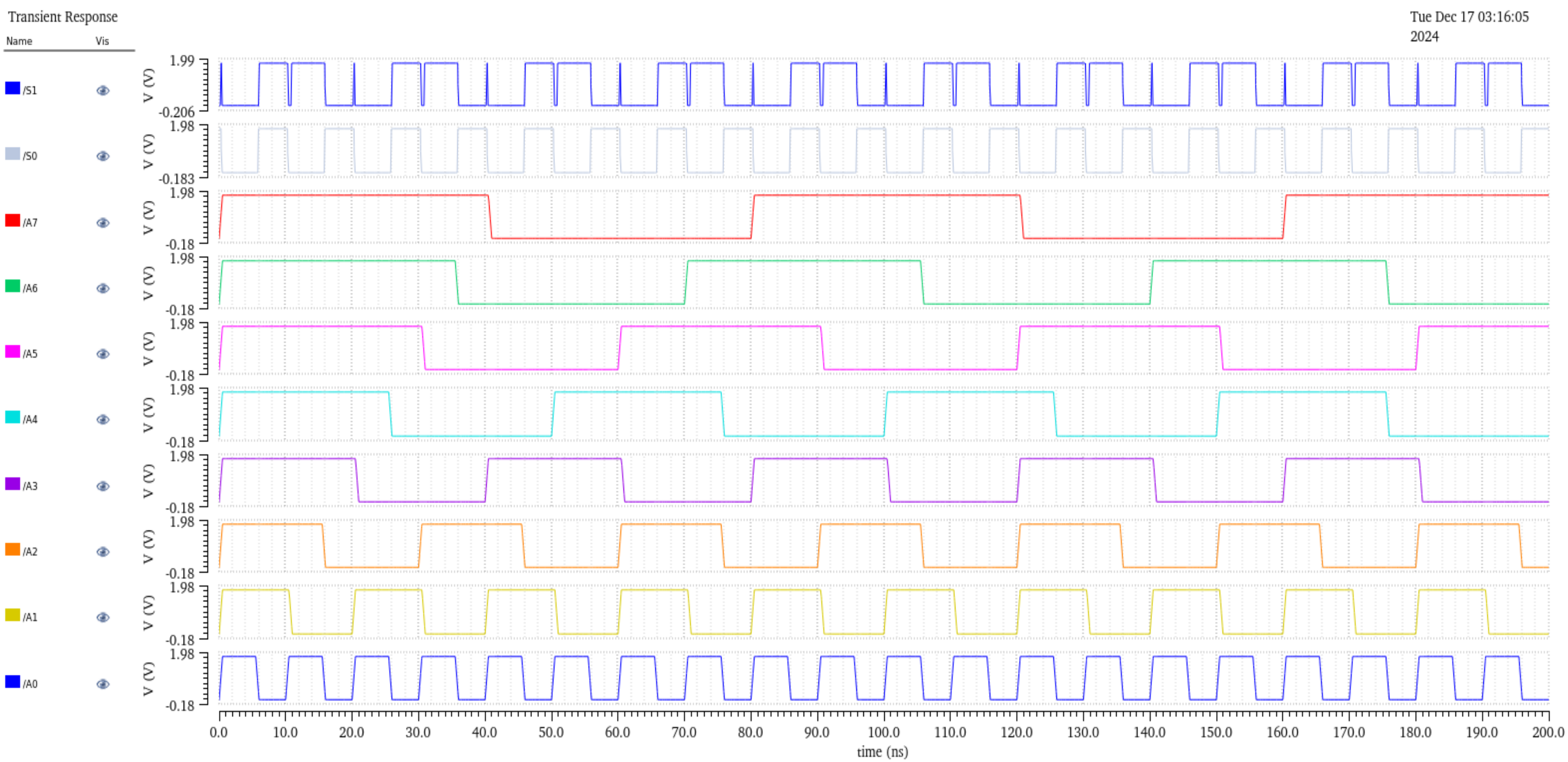
RUN WINDOW.



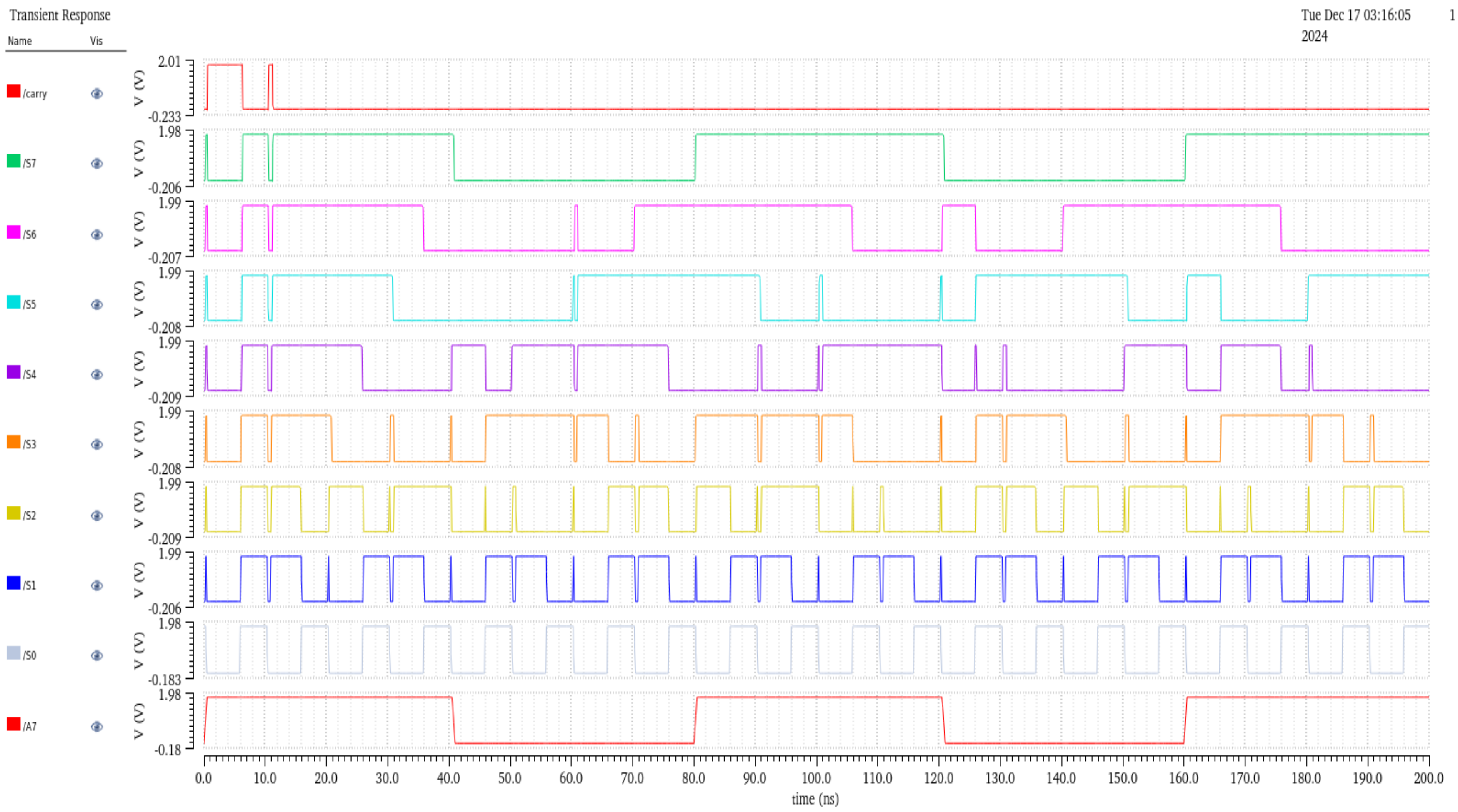
Choosing Analyses -- ADE L (2)
Analysis
tran
dc
ac
noise
xf
sens
dcmatch
acmatch
stb
pz
lf
sp
envlp
pss
pac
pstb
pnoise
pxf
psp
qpss
qpac
qpnoise
qpxf
qpsp
hb
hbac
hbstb
hbnoise
hbsp
hbxf
Transient Analysis
Stop Time
200n
Accuracy Defaults (errpreset)
conservative
moderate
liberal
Transient Noise
Dynamic Parameter
Enabled
Options...
OK
Cancel
Defaults
Apply
Help

OUTPUT FOR 8 BIT INCREMENTER.

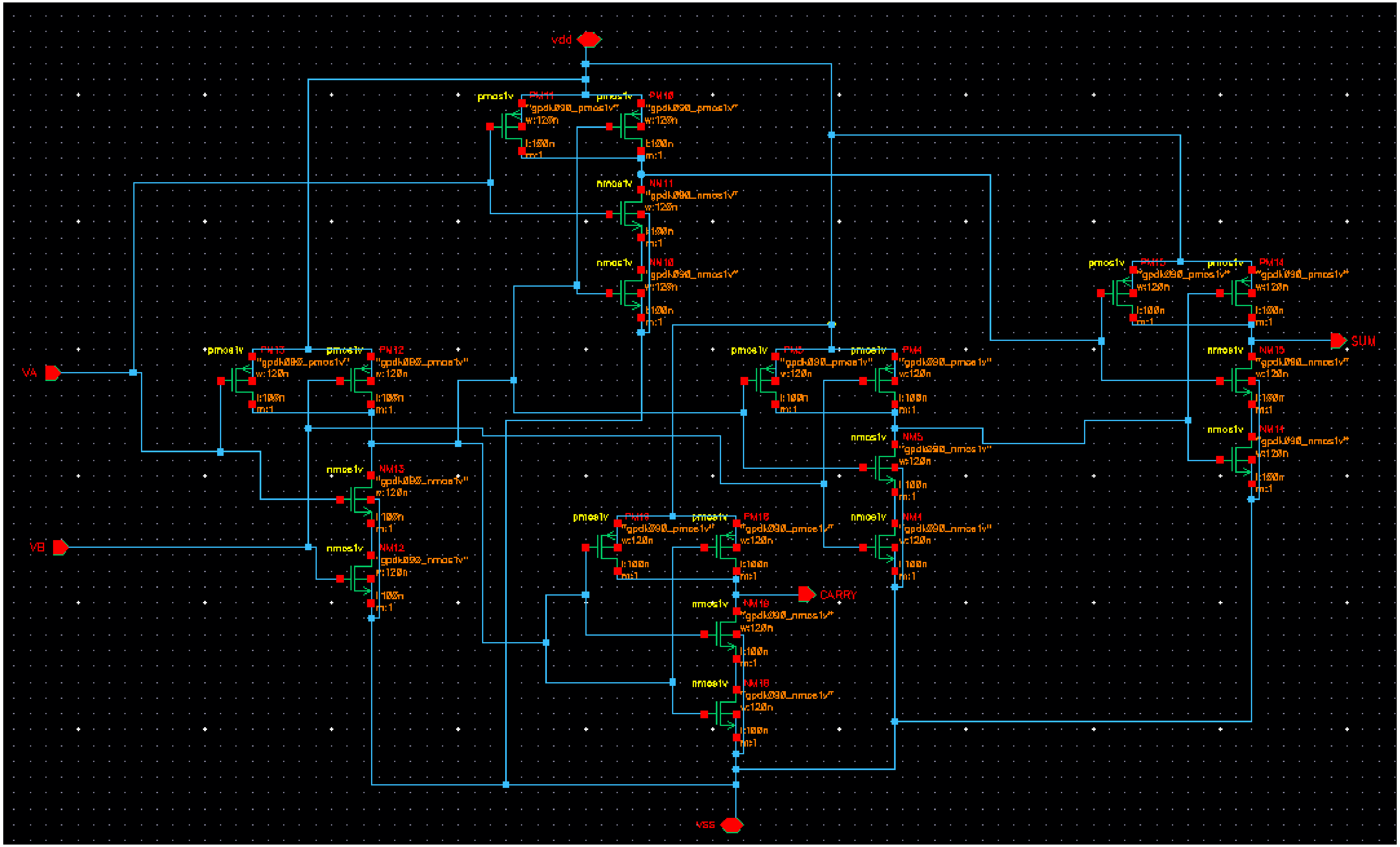
INPUT BITS.



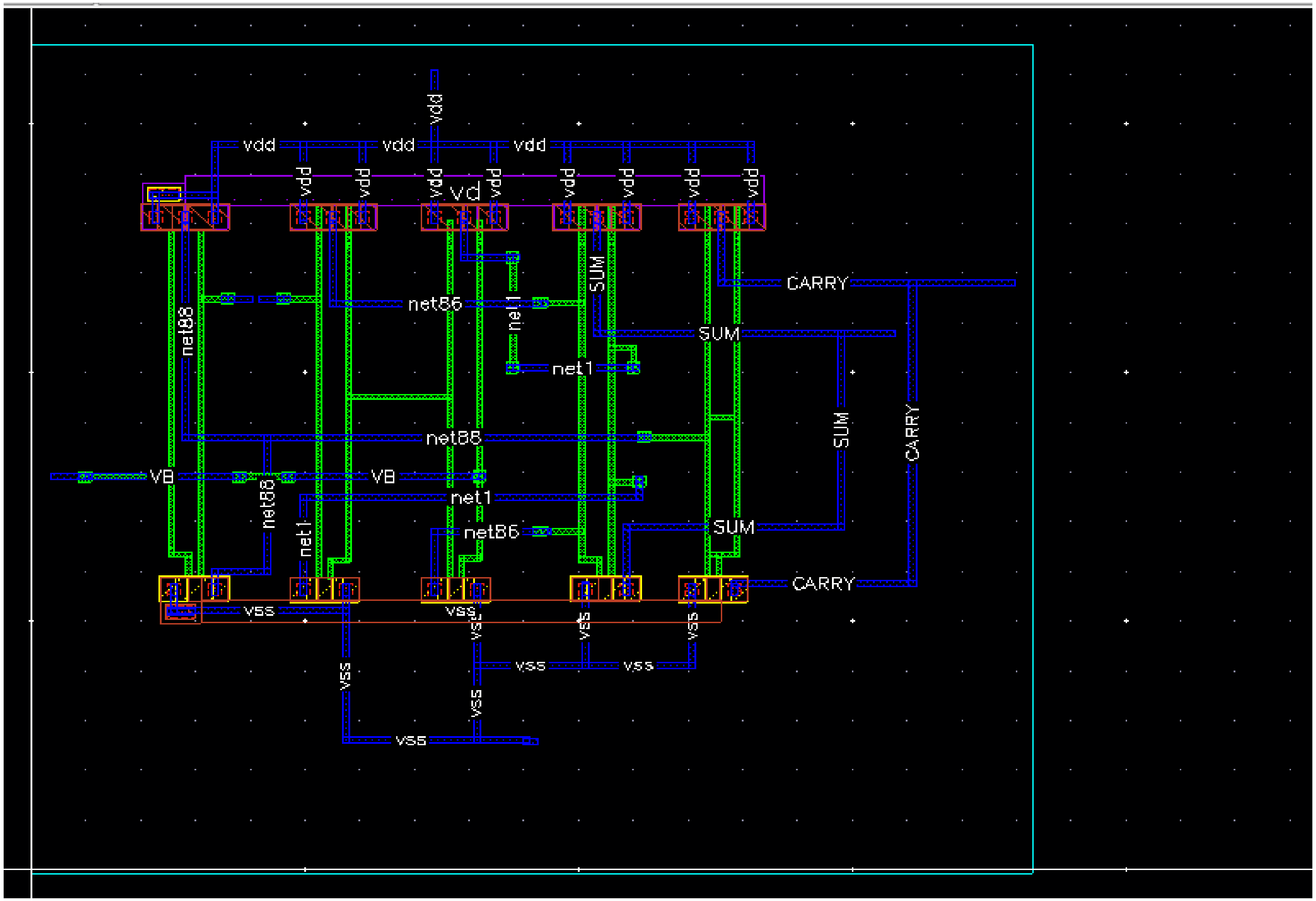
OUTPUT BITS.


Transient Response
Tue Dec 17 03:16:05 2024
1
Name
Vis
/carry
/S7
/S6
/S5
/S4
/S3
/S2
/S1
/S0
/A7
V (V)
2.01
-0.233
1.98
-0.206
1.99
-0.207
1.99
-0.208
1.99
-0.209
1.99
-0.208
1.99
-0.209
1.99
-0.206
1.98
-0.183
1.98
-0.18
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
110.0
120.0
130.0
140.0
150.0
160.0
170.0
180.0
190.0
200.0
time (ns)

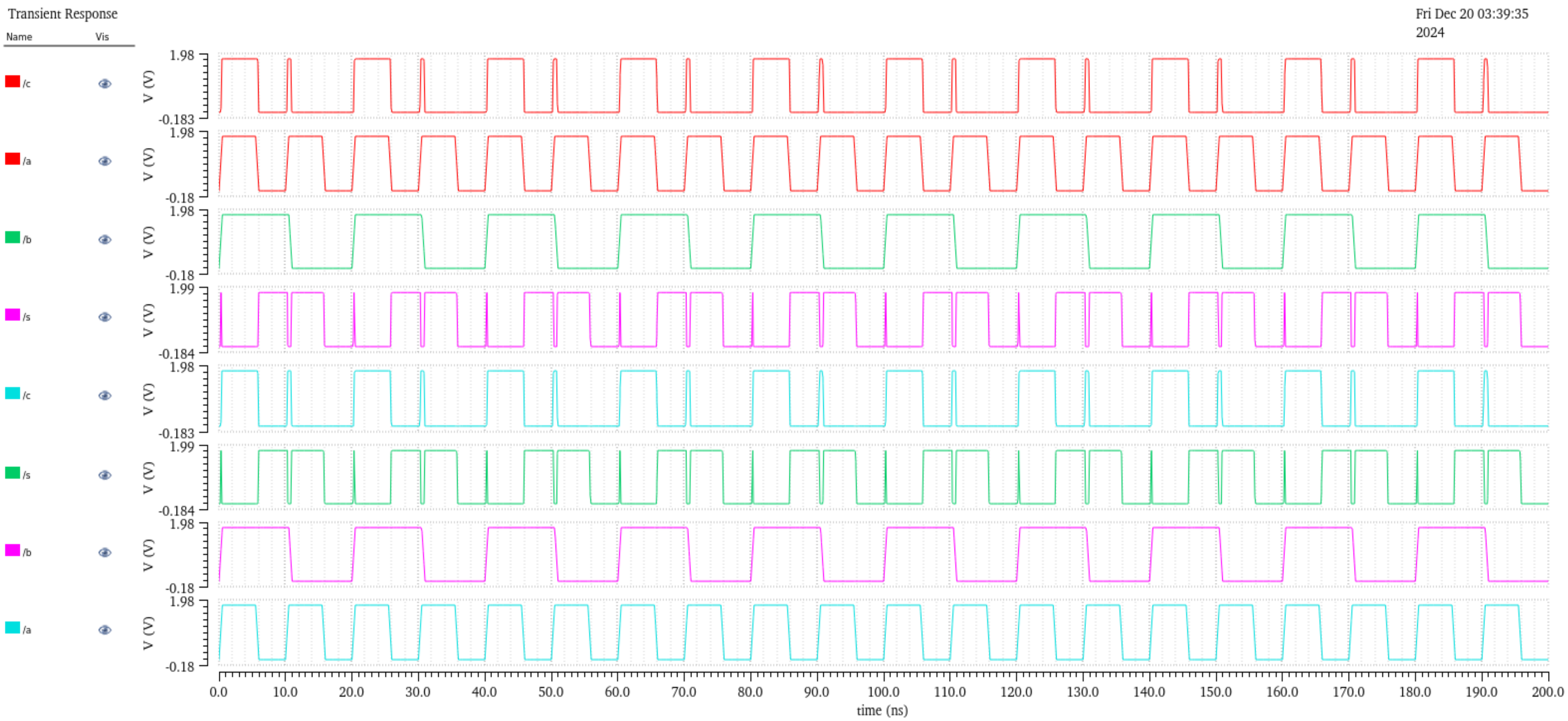
STEP 4 - CREATE A TRANSISTOR LOGIC FOR HALF ADDER USING PMOS NMOS.



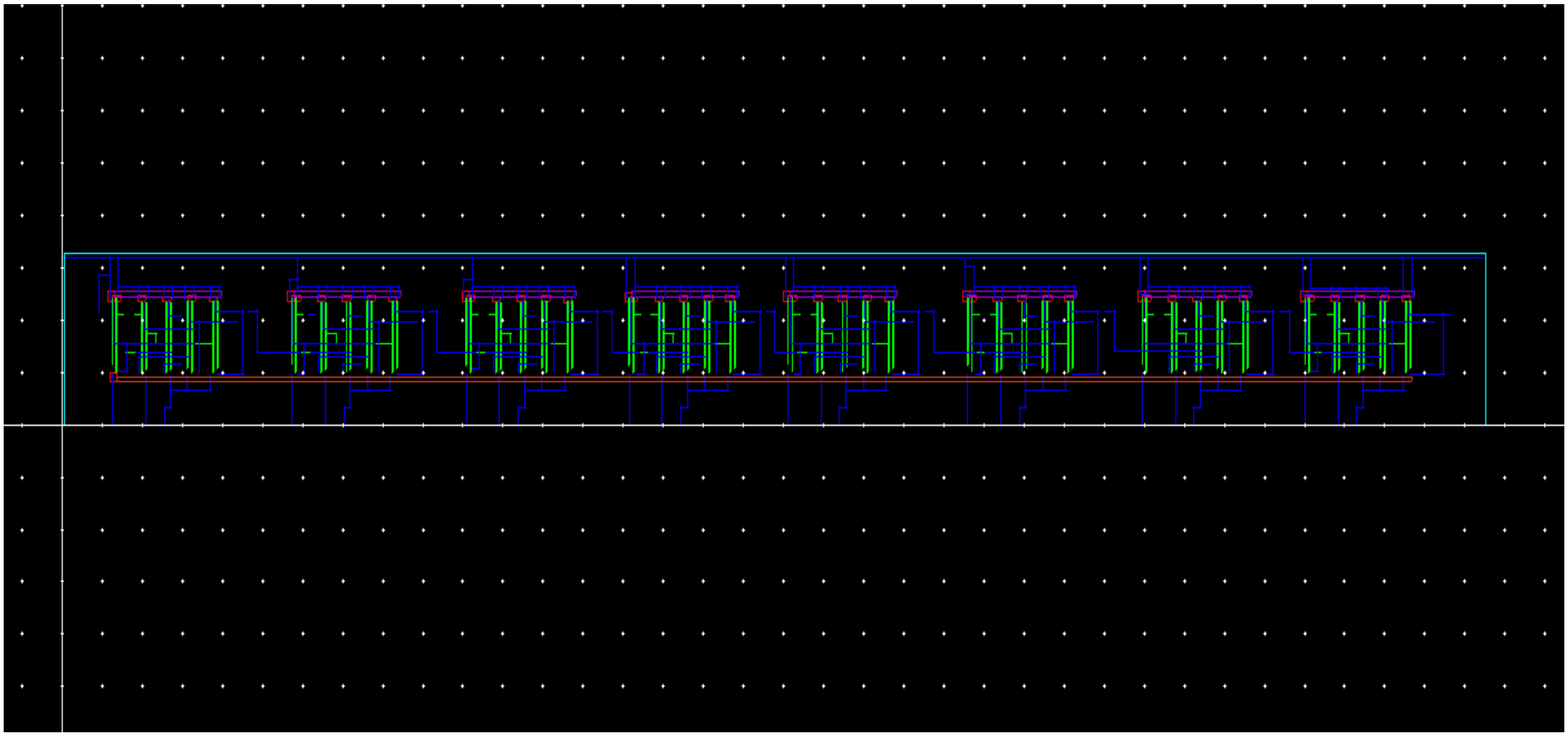
CREATE LAYOUT FOR HALF ADDER.



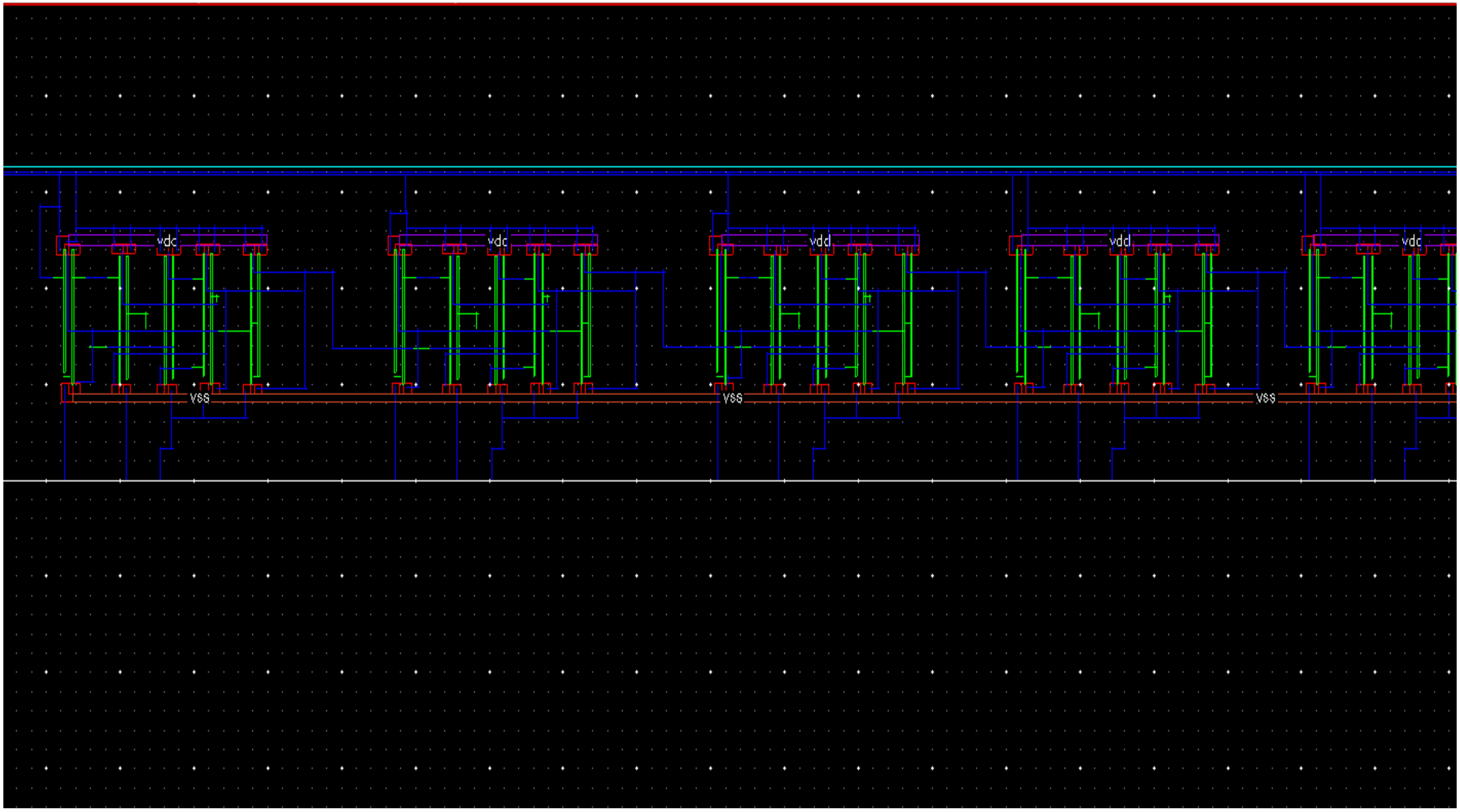
OUTPUT FOR LAYOUT OF HALF ADDER.

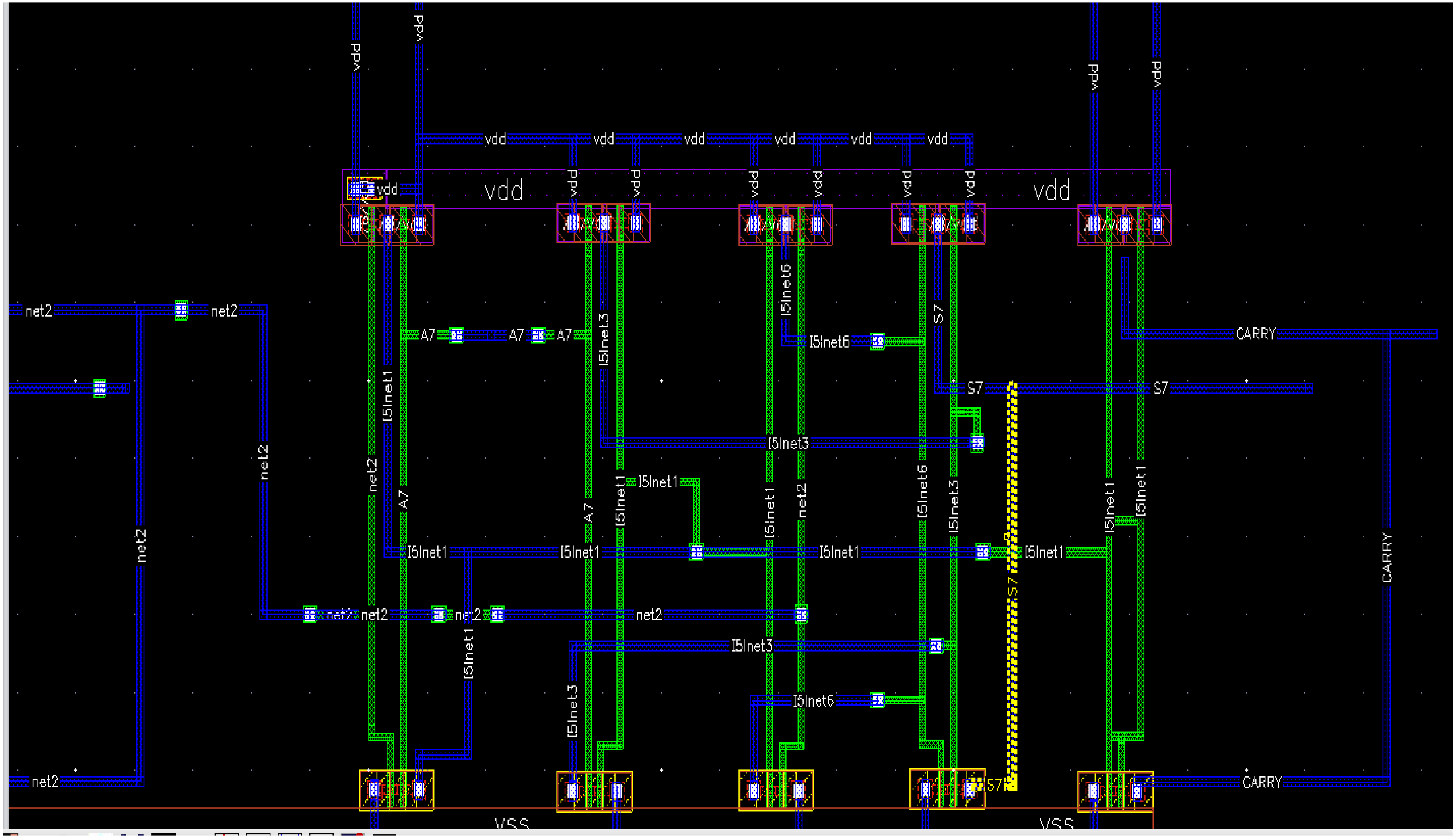


LAYOUT OF 8 BIT INCREMENTER EXTRACTED FROM 8 BIT INCREMENTER'S SCHEMATIC.



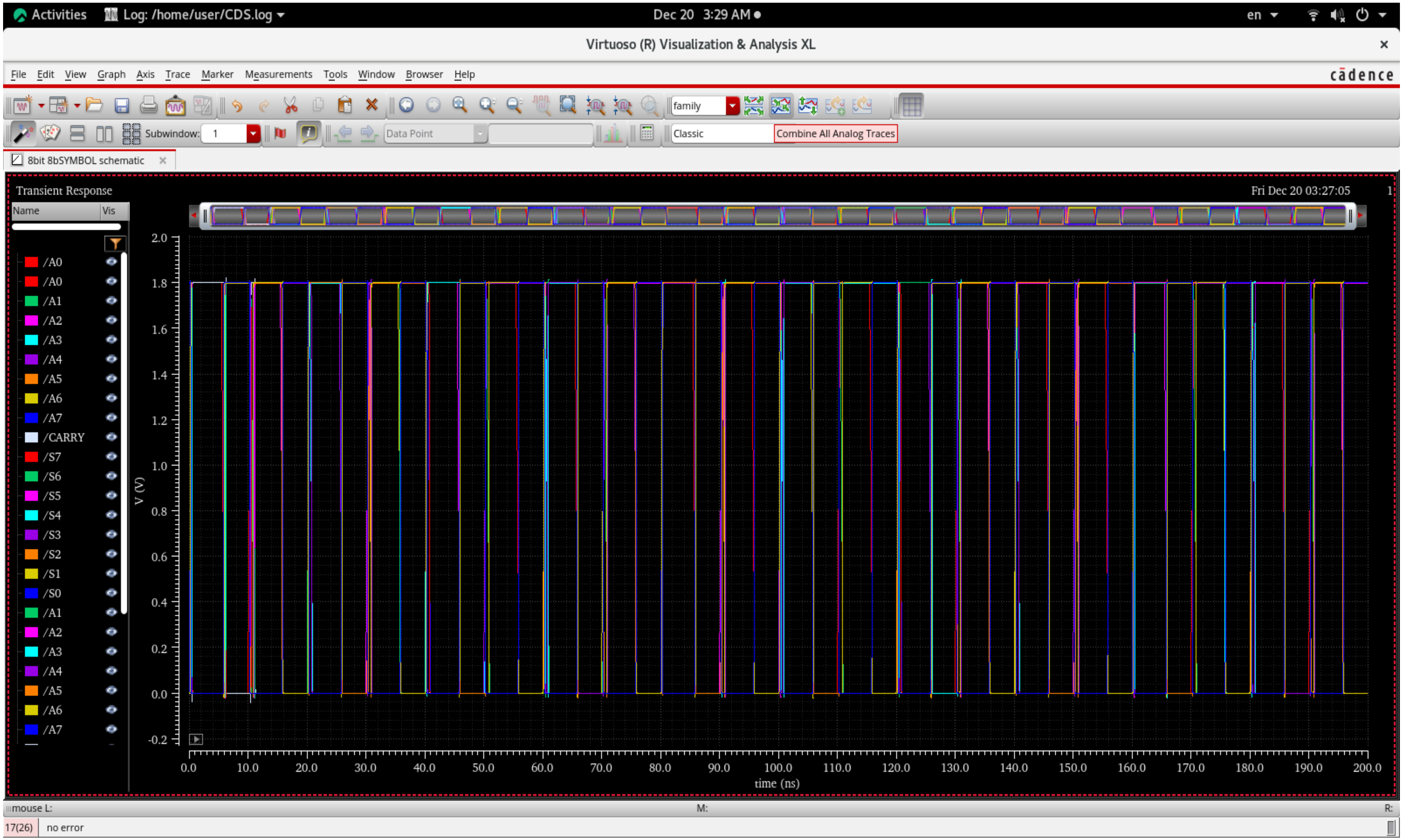
ZOOMED IN.



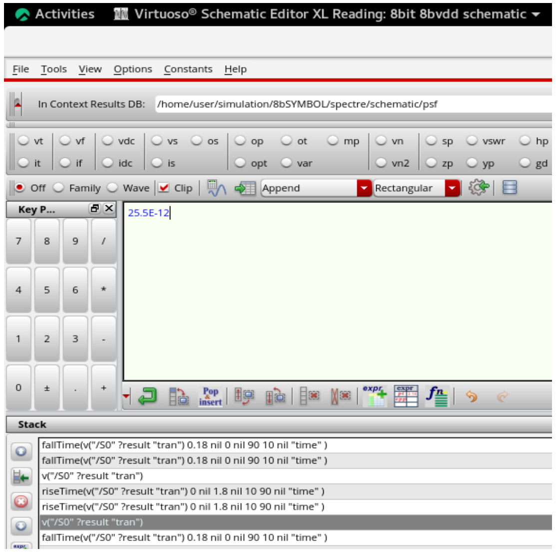


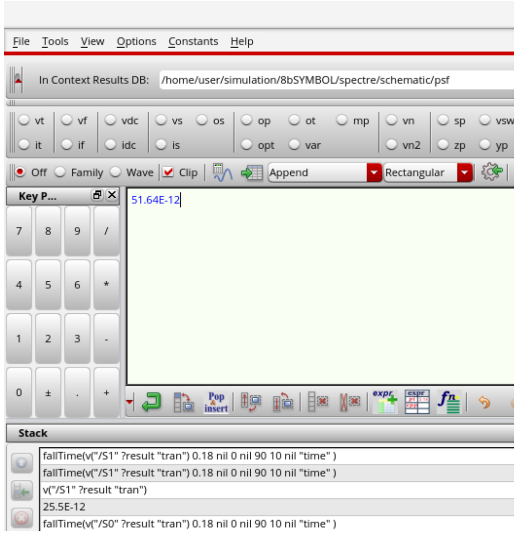
vdd
net2
A7
I5lnet1
I5lnet3
I5lnet6
S7
CARRY
VSS

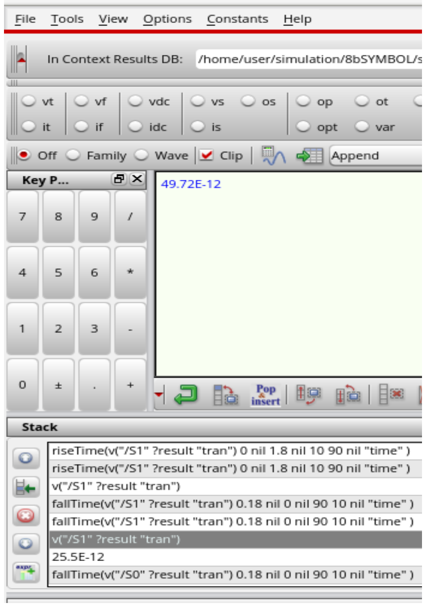
OUTPUT FOR LAYOUT OF 8 BIT INCREMENTER.

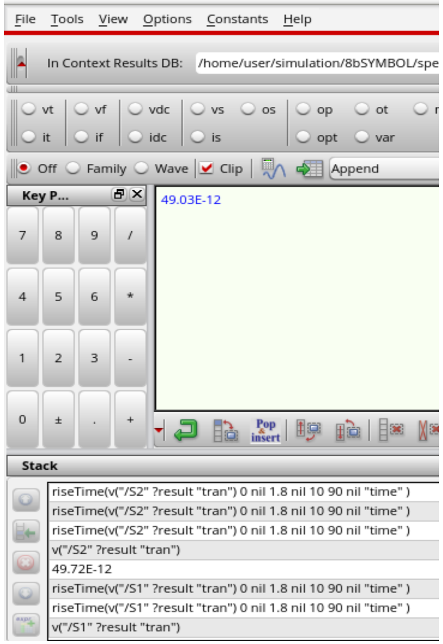


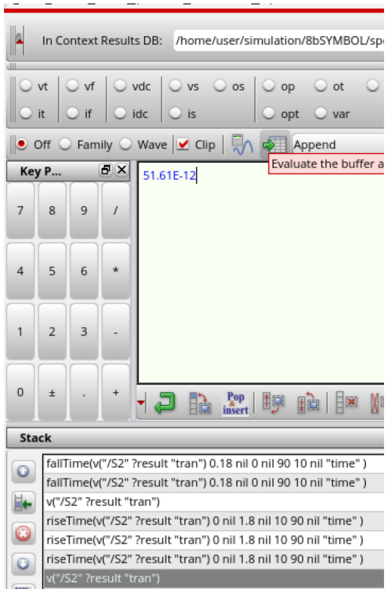
RISE AND FALL TIME.

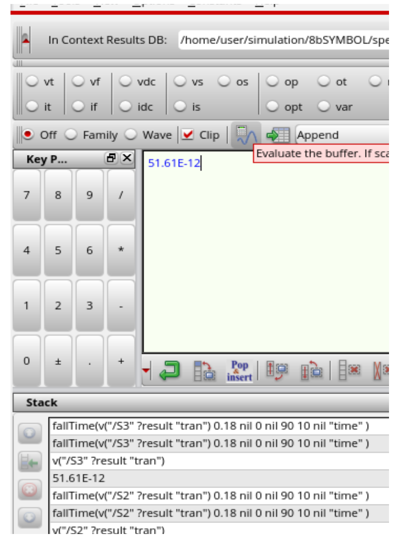


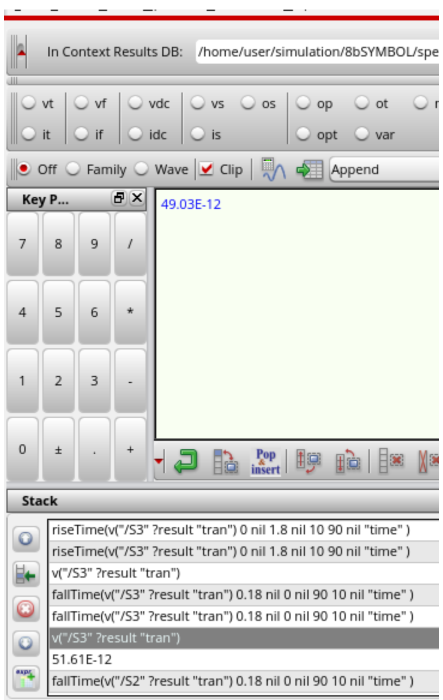




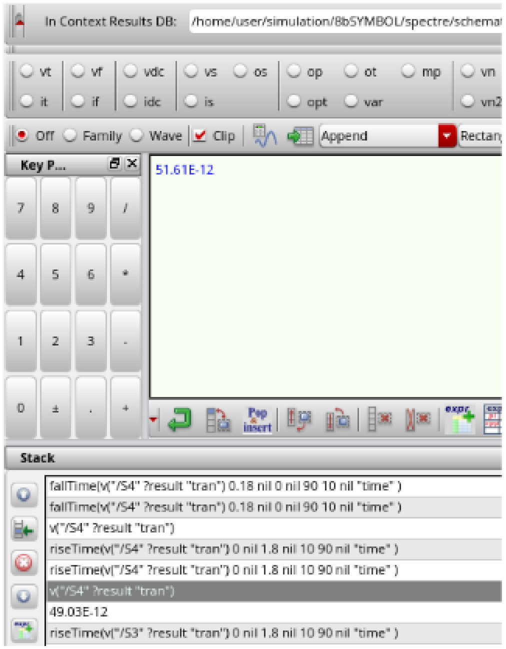


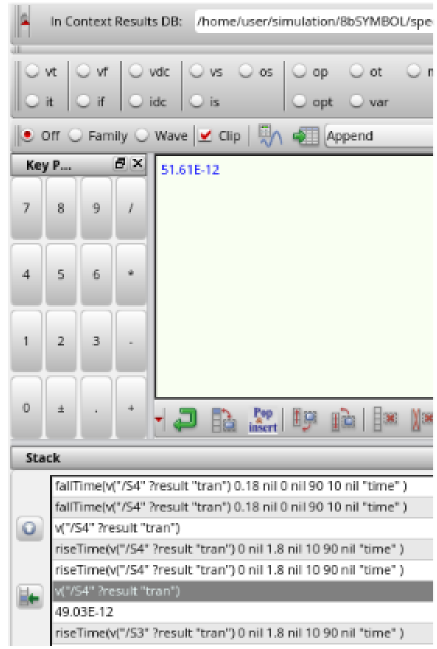

In Context Results DB: /home/user/simulation/8bSYMBOL/sp
Off Family Wave Clip Append
Evaluate the buffer a
Key P...
51.61E-12
Stack
fallTime(v("/S2" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
fallTime(v("/S2" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S2" ?result "tran")
riseTime(v("/S2" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
riseTime(v("/S2" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
riseTime(v("/S2" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S2" ?result "tran")

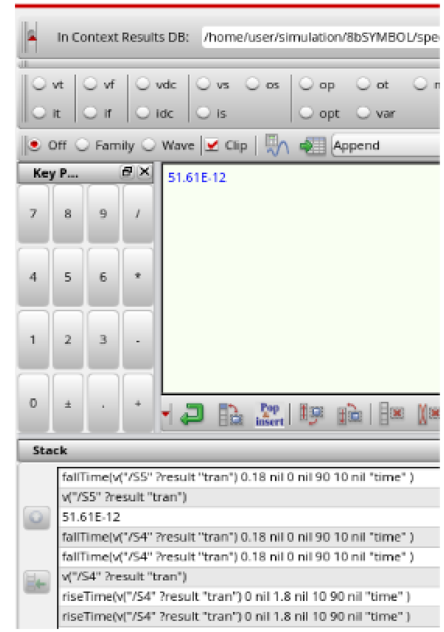

In Context Results DB: /home/user/simulation/8bSYMBOL/spe
Off Family Wave Clip Append
Evaluate the buffer. If sca
Key P...
51.61E-12
Stack
fallTime(v("/S3" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
fallTime(v("/S3" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S3" ?result "tran")
51.61E-12
fallTime(v("/S2" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
fallTime(v("/S2" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S2" ?result "tran")

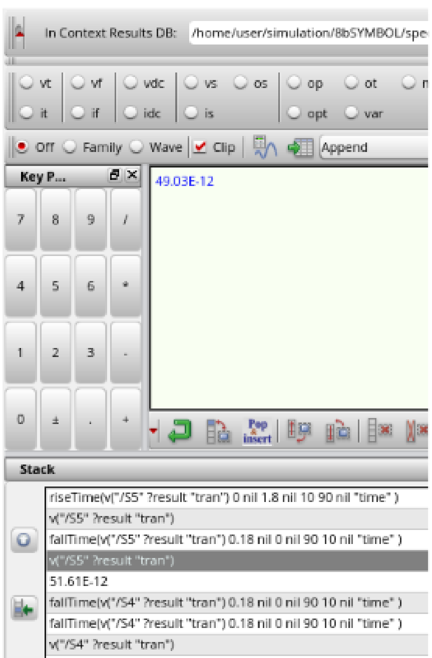

In Context Results DB: /home/user/simulation/8bSYMBOL/spe
Off Family Wave Clip Append
Key P...
49.03E-12
Stack
riseTime(v("/S3" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
riseTime(v("/S3" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S3" ?result "tran")
fallTime(v("/S3" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
fallTime(v("/S3" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S3" ?result "tran")
51.61E-12
fallTime(v("/S2" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")


File Tools View Options Constants Help
In Context Results DB: /home/user/simulation/8bSYMBOL/spectr
Off Family Wave Clip Append
Key P...
49.03E-12
Stack
riseTime(v("/S4" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
riseTime(v("/S4" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S4" ?result "tran")
49.03E-12
riseTime(v("/S3" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
riseTime(v("/S3" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S3" ?result "tran")
fallTime(v("/S3" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")

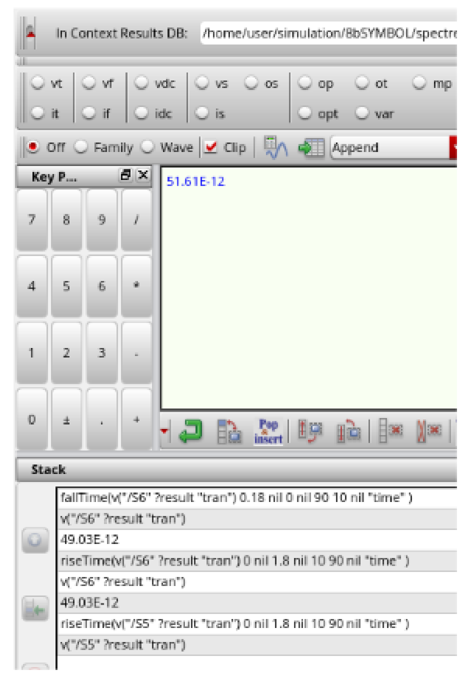

In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schemat
51.61E-12
Stack
fallTime(v("/S4" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
fallTime(v("/S4" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S4" ?result "tran")
riseTime(v("/S4" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
riseTime(v("/S4" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S4" ?result "tran")
49.03E-12
riseTime(v("/S3" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")

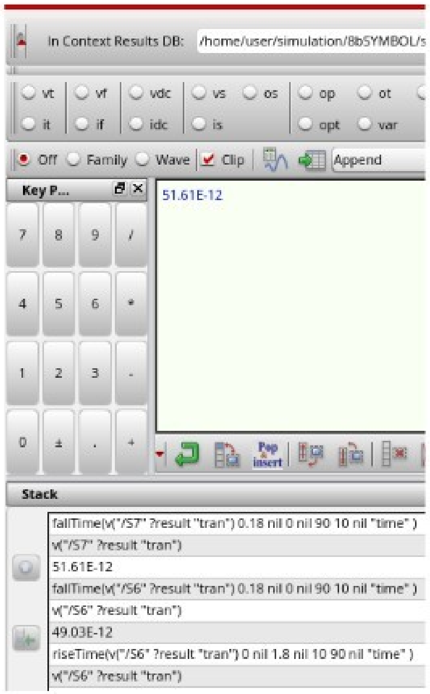

In Context Results DB: /home/user/simulation/8bSYMBOL/spe
51.61E-12
Stack
fallTime(v("/S4" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
fallTime(v("/S4" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S4" ?result "tran")
riseTime(v("/S4" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
riseTime(v("/S4" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S4" ?result "tran")
49.03E-12
riseTime(v("/S3" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")

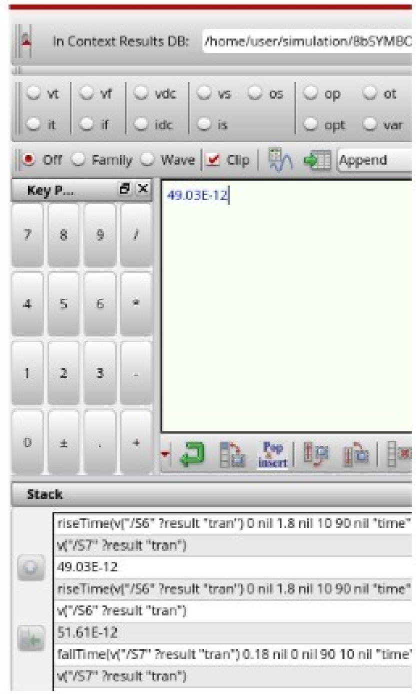

In Context Results DB: /home/user/simulation/8bSYMBOL/spe
51.61E-12
Stack
fallTime(v("/S5" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S5" ?result "tran")
51.61E-12
fallTime(v("/S4" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
fallTime(v("/S4" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S4" ?result "tran")
riseTime(v("/S4" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
riseTime(v("/S4" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")

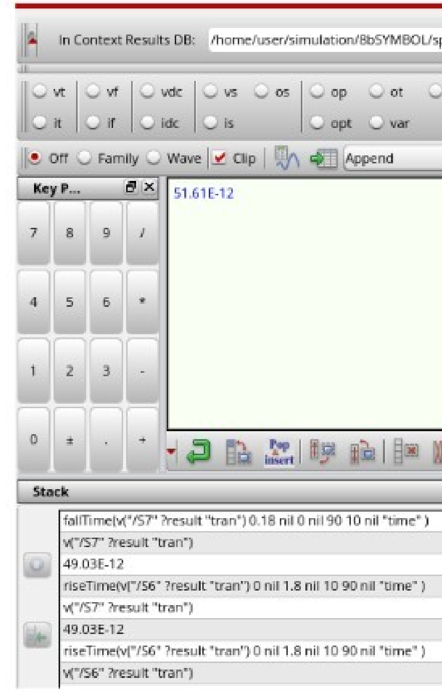

In Context Results DB: /home/user/simulation/8bSYMBOL/spe
49.03E-12
Stack
riseTime(v("/S5" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S5" ?result "tran")
fallTime(v("/S5" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S5" ?result "tran")
51.61E-12
fallTime(v("/S4" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
fallTime(v("/S4" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S4" ?result "tran")


In Context Results DB: /home/user/simulation/8bSYMBOL/spe
49.03E-12
Stack
riseTime(v("/S6" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S6" ?result "tran")
49.03E-12
riseTime(v("/S5" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S5" ?result "tran")
fallTime(v("/S5" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S5" ?result "tran")
51.61E-12

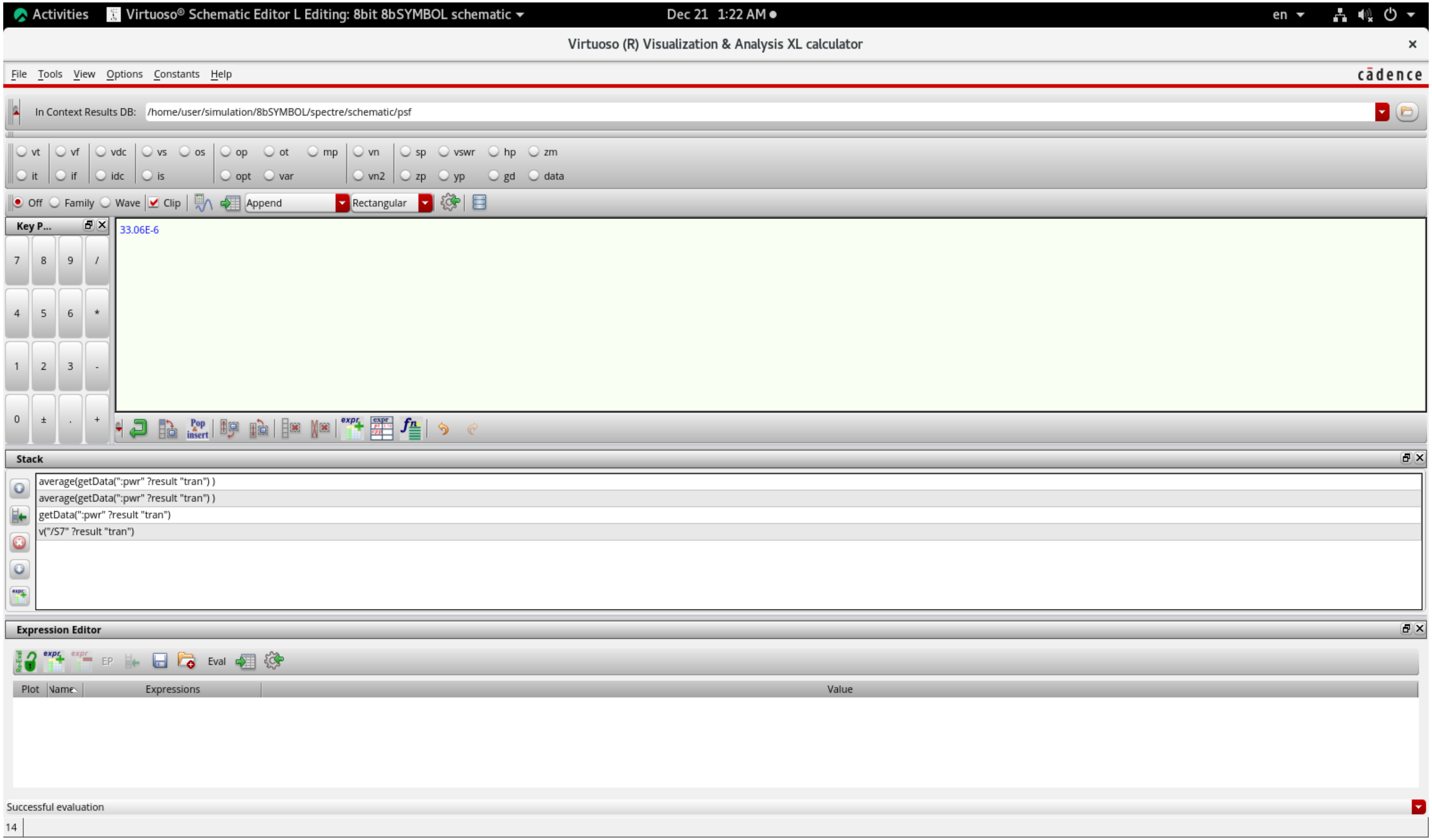

In Context Results DB: /home/user/simulation/8bSYMBOL/spectre
51.61E-12
Stack
fallTime(v("/S6" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S6" ?result "tran")
49.03E-12
riseTime(v("/S6" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S6" ?result "tran")
49.03E-12
riseTime(v("/S5" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S5" ?result "tran")

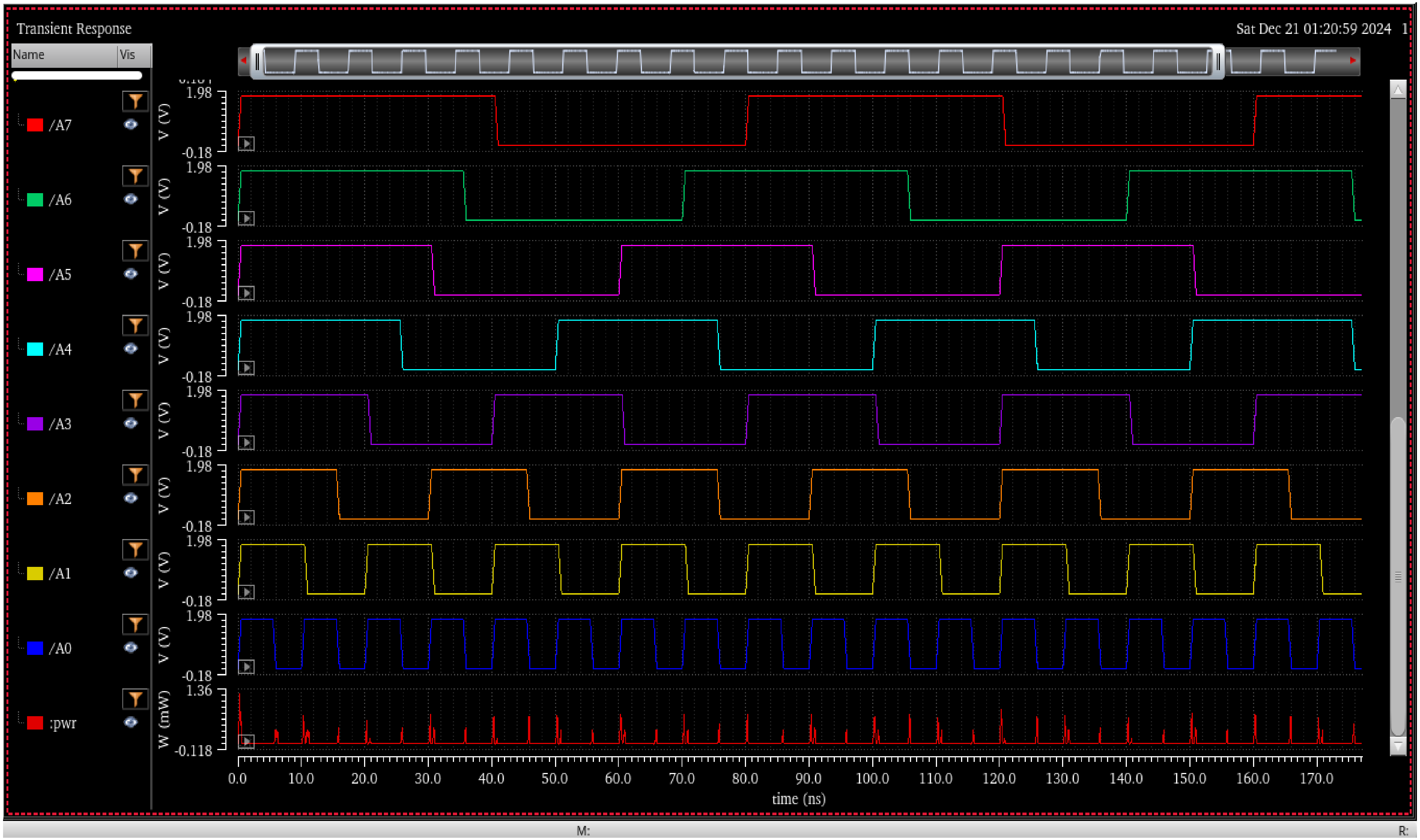

In Context Results DB:
/home/user/simulation/8bSYMBOL/s
vt
vf
vdc
vs
os
op
ot
it
if
idc
is
opt
var
Off
Family
Wave
Clip
Append
Key P...
51.61E-12
7
8
9
/
4
5
6
*
1
2
3
-
0
±
.
+
Pop insert
Stack
fallTime(v("/S7" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S7" ?result "tran")
51.61E-12
fallTime(v("/S6" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S6" ?result "tran")
49.03E-12
riseTime(v("/S6" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S6" ?result "tran")


In Context Results DB:
/home/user/simulation/8bSYMBO
vt
vf
vdc
vs
os
op
ot
it
if
idc
is
opt
var
Off
Family
Wave
Clip
Append
Key P...
49.03E-12
7
8
9
/
4
5
6
*
1
2
3
-
0
±
.
+
Pop insert
Stack
riseTime(v("/S6" ?result "tran") 0 nil 1.8 nil 10 90 nil "time"
v("/S7" ?result "tran")
49.03E-12
riseTime(v("/S6" ?result "tran") 0 nil 1.8 nil 10 90 nil "time"
v("/S6" ?result "tran")
51.61E-12
fallTime(v("/S7" ?result "tran") 0.18 nil 0 nil 90 10 nil "time"
v("/S7" ?result "tran")

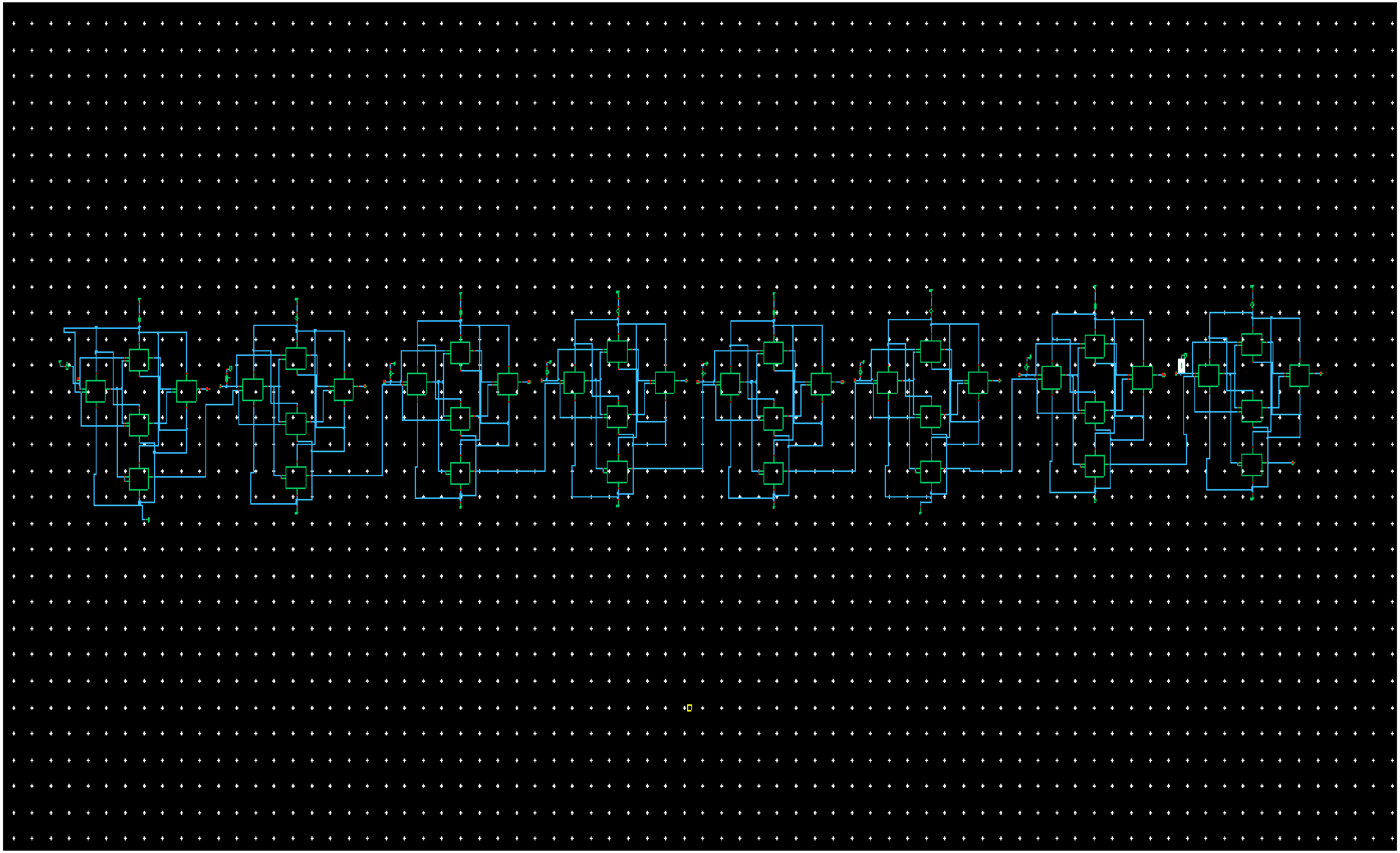

In Context Results DB:
/home/user/simulation/8bSYMBOL/spe
vt
vf
vdc
vs
os
op
ot
it
if
idc
is
opt
var
Off
Family
Wave
Clip
Append
Key P...
51.61E-12
7
8
9
/
4
5
6
*
1
2
3
-
0
±
.
+
Pop insert
Stack
fallTime(v("/S7" ?result "tran") 0.18 nil 0 nil 90 10 nil "time")
v("/S7" ?result "tran")
49.03E-12
riseTime(v("/S6" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S7" ?result "tran")
49.03E-12
riseTime(v("/S6" ?result "tran") 0 nil 1.8 nil 10 90 nil "time")
v("/S6" ?result "tran")

POWER DISSIPATION.

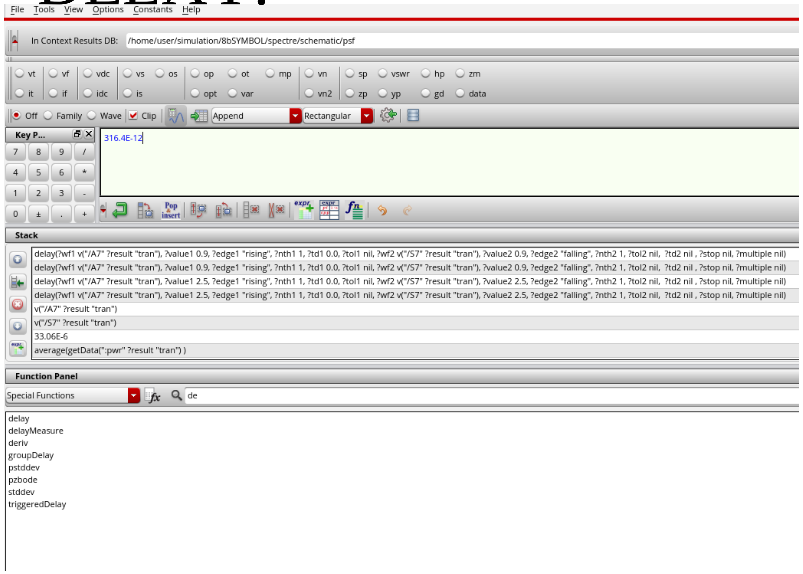


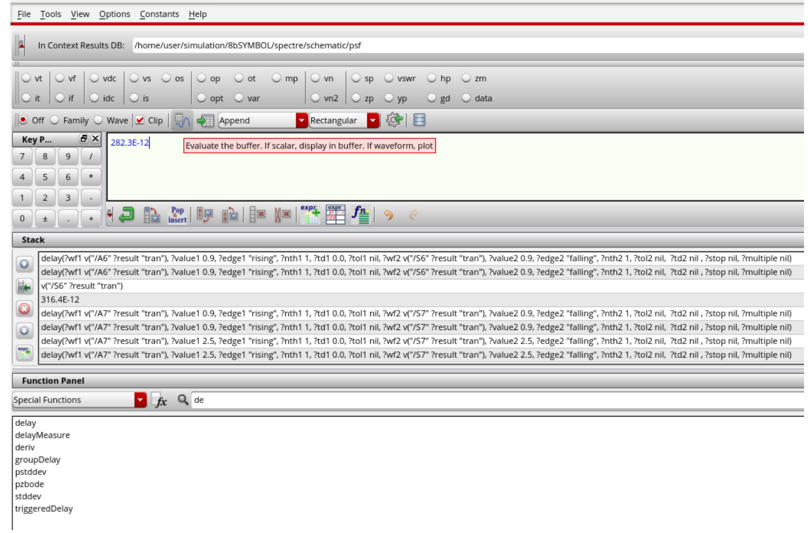


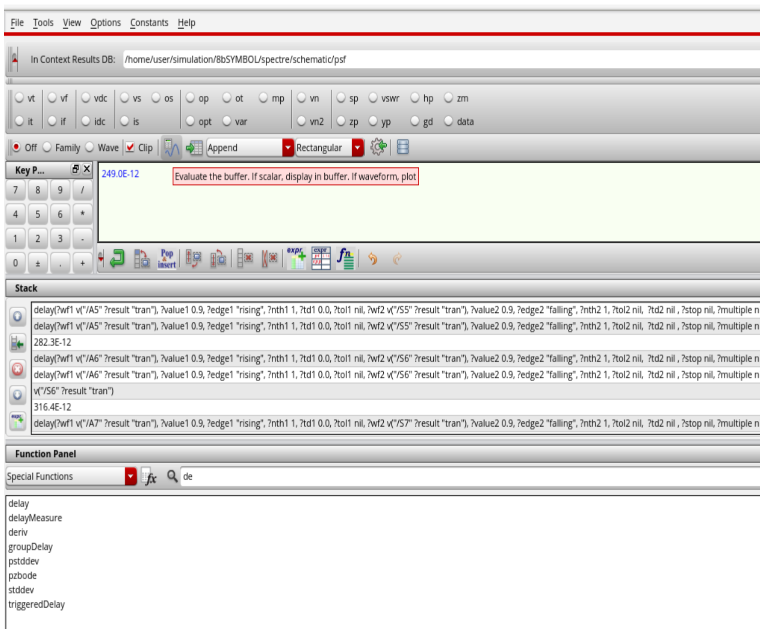
Transient Response
Sat Dec 21 01:20:59 2024
Name
Vis
/A7
/A6
/A5
/A4
/A3
/A2
/A1
/A0
:pwr
V (V)
W (mW)
1.98
-0.18
1.36
-0.118
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
110.0
120.0
130.0
140.0
150.0
160.0
170.0
time (ns)
M:
R:

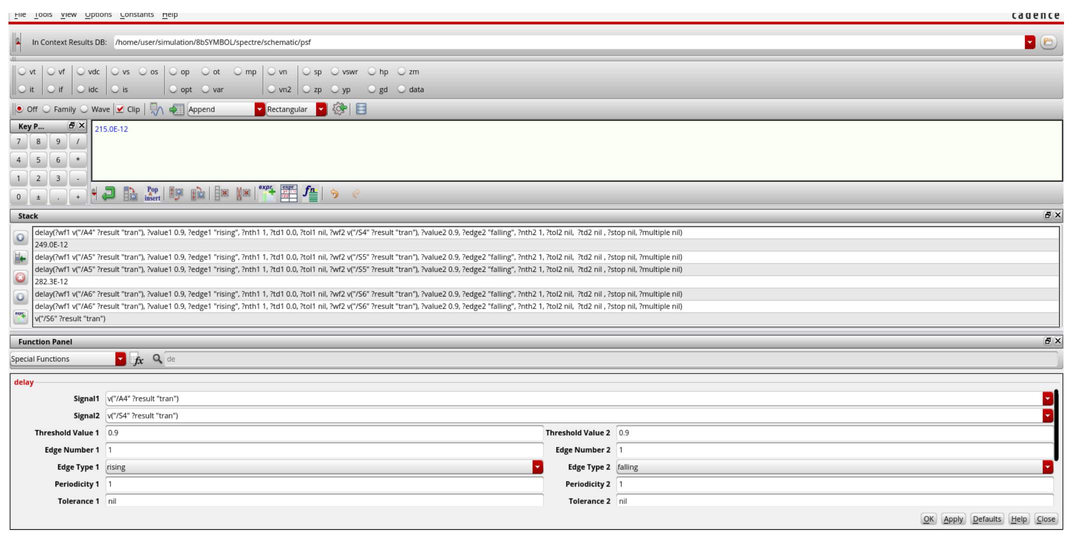


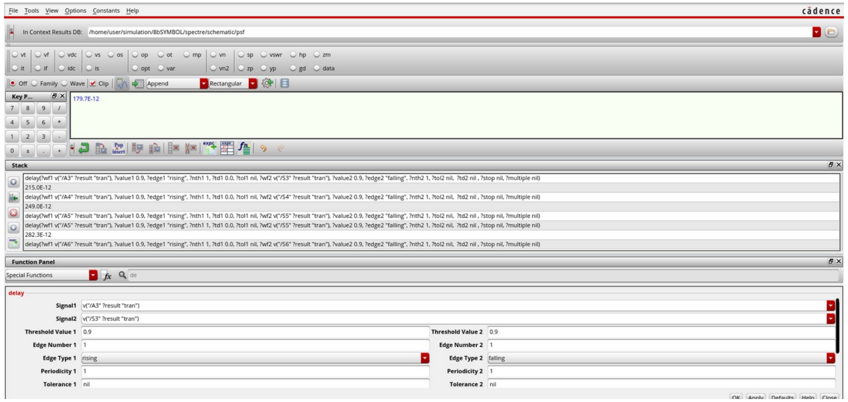
DELAY.

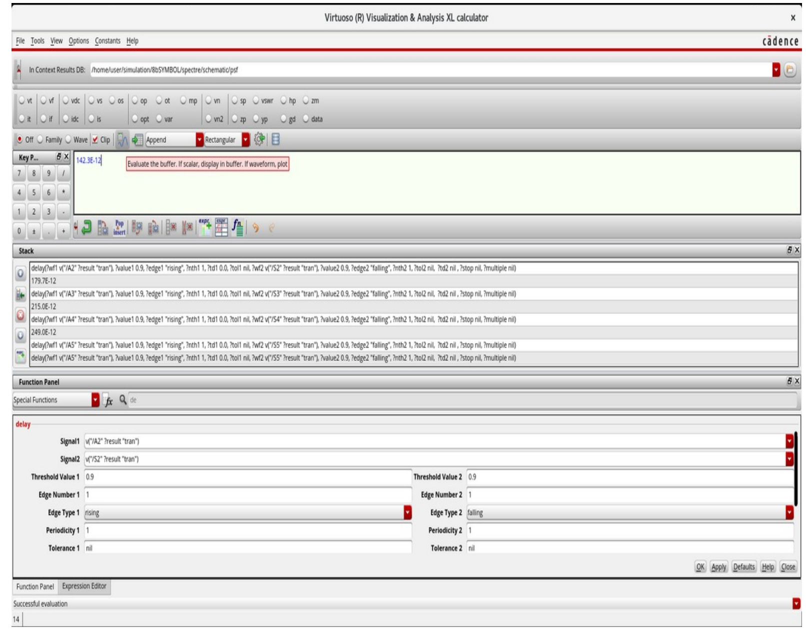

In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schematic/psf
316.4E-12
Stack
Function Panel
Special Functions
delay
delayMeasure
deriv
groupDelay
pstddev
pzbode
stddev
triggeredDelay

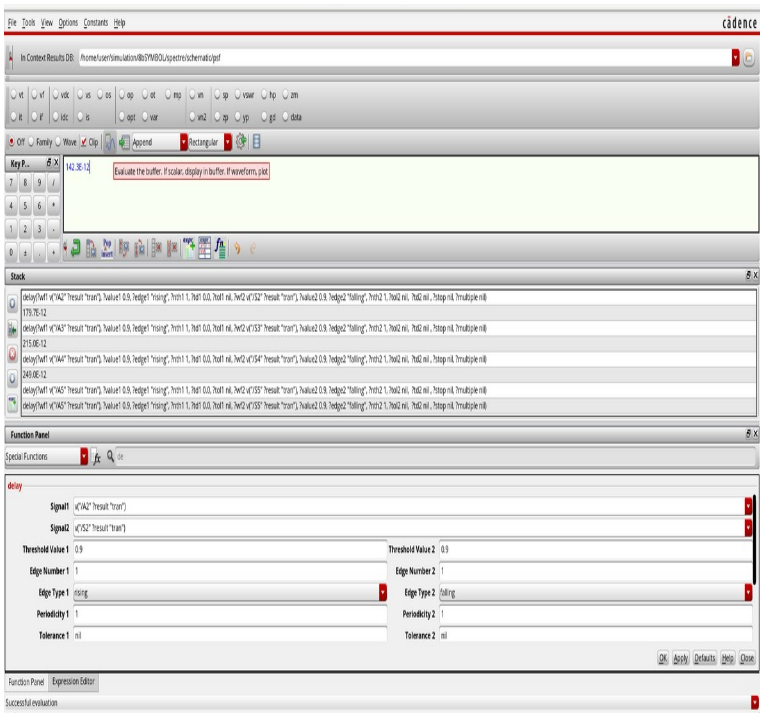

In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schematic/psf
282.3E-12
Evaluate the buffer. If scalar, display in buffer. If waveform, plot
Stack
Function Panel
Special Functions

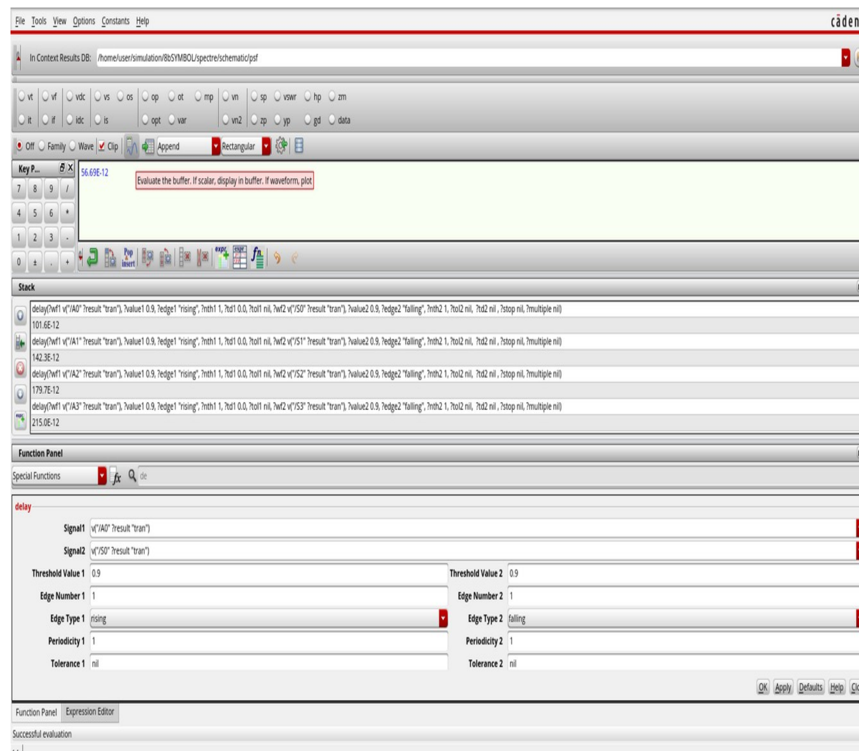

In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schematic/psf
249.0E-12
Evaluate the buffer. If scalar, display in buffer. If waveform, plot
Stack
Function Panel
Special Functions


In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schematic/psf
215.0E-12
Stack
Function Panel
delay
Signal1 v("/A4" ?result "tran")
Signal2 v("/S4" ?result "tran")
Threshold Value 1 0.9
Threshold Value 2 0.9
Edge Number 1 1
Edge Number 2 1
Edge Type 1 rising
Edge Type 2 falling
Periodicity 1 1
Periodicity 2 1
Tolerance 1 nil
Tolerance 2 nil
OK Apply Defaults Help Close


File Tools View Options Constants Help
In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schematic/psf
179.7E-12
Stack
Function Panel
Special Functions
delay


Virtuoso (R) Visualization & Analysis XL calculator
File Tools View Options Constants Help
In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schematic/psf
142.3E-12
Evaluate the buffer. If scalar, display in buffer. If waveform, plot
Stack
Function Panel
Special Functions
delay
Function Panel
Expression Editor
Successful evaluation


File Tools View Options Constants Help
In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schematic/psf
142.3E-12
Evaluate the buffer. If scalar, display in buffer. If waveform, plot
Stack
Function Panel
Special Functions
delay
Function Panel
Expression Editor
Successful evaluation


File Tools View Options Constants Help
In Context Results DB: /home/user/simulation/8bSYMBOL/spectre/schematic/psf
56.69E-12
Evaluate the buffer. If scalar, display in buffer. If waveform, plot
Stack
Function Panel
Special Functions
delay
Function Panel
Expression Editor
Successful evaluation